



AHLT Lab Project Report

Task 1 — NERC (Named Entity Recognition and Classification)

Contents

1	Introduction	2
1.1	Context and objective	2
1.2	Task 1 deliverables	2
2	Data, evaluation, and reproducibility	2
2.1	Dataset	2
2.2	Evaluation protocol	2
2.3	Reproducibility	2
3	System 1.1 — NERC with Machine Learning (done)	3
3.1	Baseline	3
3.2	Feature engineering experiments	3
3.3	Algorithm comparison (CRF, MEM, SVM)	3
3.4	Hyperparameter experiments	3
3.5	Best obtained results and discussion	3
4	System 1.2 — NERC with Neural Networks (pending)	3
5	System 1.3 — NERC with LLMs (pending)	3
6	Conclusions (pending)	3

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10

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1 Introduction

1.1 Context and objective

This project is based on SemEval-2013 Task 9 (biomedical information extraction). Task 1 is **Named Entity Recognition and Classification (NERC)**: identify drug mentions and classify them into the provided entity types.

1.2 Task 1 deliverables

The Task 1 report requires documenting experiments and best results for:

- System 1.1: NERC with feature-based Machine Learning (features, algorithms, hyperparameters).
- System 1.2: NERC with Neural Networks (input representations, architecture variants, hyperparameters).
- System 1.3: NERC with LLMs (prompt adaptation, few-shot, fine-tuning).

and a final comparison (effort, computational cost, explainability/controllability, and performance).

2 Data, evaluation, and reproducibility

2.1 Dataset

Briefly describe train/devel/test splits and label set.

2.2 Evaluation protocol

Briefly describe the official evaluator and the reported metrics.

2.3 Reproducibility

Summarize how to run the pipeline and where results are stored (e.g., `code/1.1.NERC-ML/results/` and `code/1.1.NERC-ML/results_c_change/`).

3 System 1.1 — NERC with Machine Learning (done)

3.1 Baseline

3.2 Feature engineering experiments

3.3 Algorithm comparison (CRF, MEM, SVM)

3.4 Hyperparameter experiments

3.5 Best obtained results and discussion

4 System 1.2 — NERC with Neural Networks (pending)

This section will be completed once the NN system experiments are done.

5 System 1.3 — NERC with LLMs (pending)

This section will be completed once the LLM prompt/few-shot/fine-tuning experiments are done.

6 Conclusions (pending)

Final comparison across 1.1/1.2/1.3 will be completed once all systems are implemented.